2016 — (Hint Bonus Nu Neville)

Extrap. R.

1. $M = N_1(h) + k_1 h + k_2 h^2 + \dots$

valabilă pt orice $h > 0$

$\Rightarrow h/2$ formulă $O(h)$

$$M = N_1\left(\frac{h}{2}\right) + k_1 \frac{h}{2} + k_2 \frac{h^2}{4} + \dots$$

$$M = N_1\left(\frac{h}{2}\right) + \left[N_1\left(\frac{h}{2}\right) - N_1(h)\right] + k_2\left(\frac{h^2}{2} - h^2\right) + k_3\left(\frac{h^3}{4} - h^3\right) + \dots$$

$\Rightarrow N_2(h) = N_1\left(\frac{h}{2}\right) + \left[N_1\left(\frac{h}{2}\right) - N_1(h)\right]$

$\Rightarrow M = N_2(h) - \frac{k_2}{2} h^2 - \frac{3k_3}{4} h^3 - \dots$

formula $O(h^2)$

2. ✓
3. ✓
4. ✓
5. $y' = y - t^2 + 1, \quad t \in [0,2]$

$y(0) = 0.5$

Euler $h = 0.5$

$y(1.3) = ?$ 1.5 ✓

$y(0,5) = y(0) + f(0, y(0)) * h$

$y(0,5) = 0.5 + 1.5 \cdot 0,5$

$= 2,5 \cdot 0,5 = 1,25$

$y(1) = y(0,5) + f(0,5, y(0,5)) \cdot h$

$y(1) = 1,25 + (1,25 - 0,25 + 1) \cdot 0,5$

$= 1,25 + 2 \cdot 0,5 = 2,25$

$y(1,5) = y(1) + f(1, y(1)) \cdot h$

$= 2,25 + (2,25 - 1 + 1) \cdot 0,5$

$= 2,25 + 1,125 = 3,375$

6. Curs 12

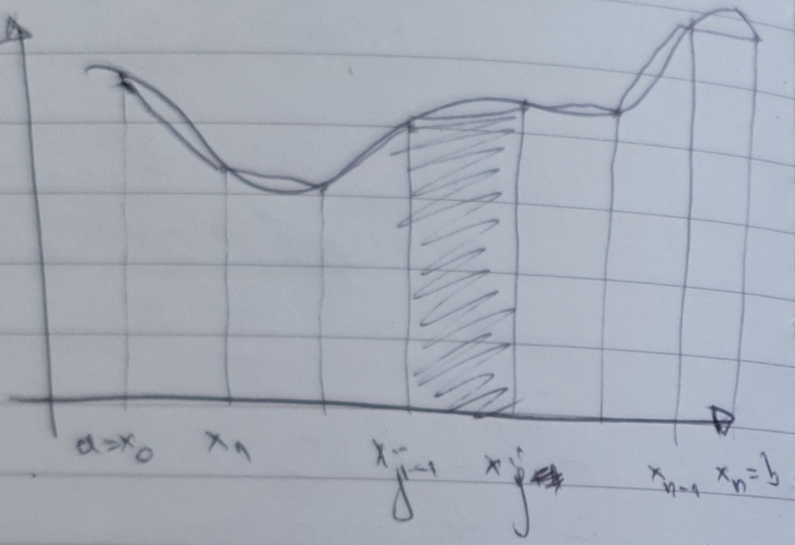
7.

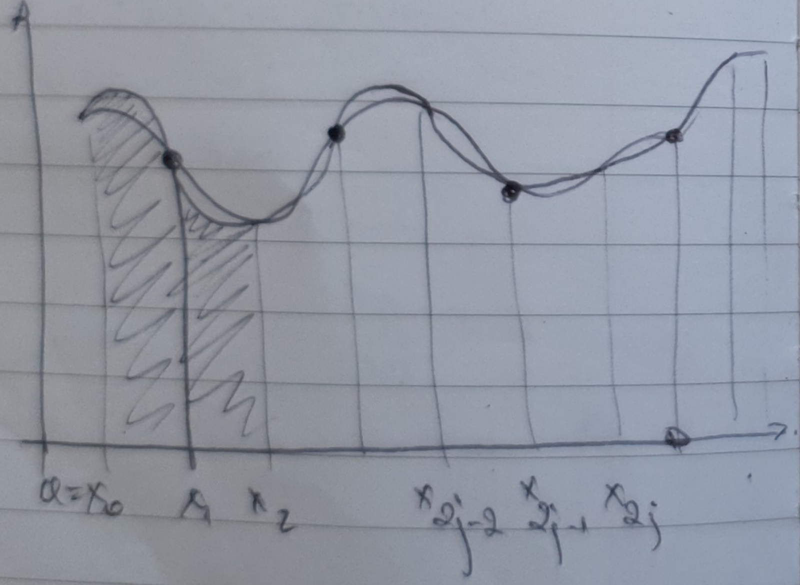
```
function trapeze_compuse(a, b, N, f)
    rez = f(a) + f(b); h = $(b-a)/N$;
    for i = 1:N-1
        rez += 2*f(a+i*h);
    endfor

    rez = rez*h/2;
endfunction
```

8. ✓

2016

1.





2. ✓
3. ✓
4. ?
5. ✓
6. ✓
7. ✓
8. ✓

$x_0, \quad x_1 = x_0 + h$

$$f(x) = L_1(x) + \frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x))$$

$$L_1(x) = f(x_0)\ell_0(x) + f(x_0+h)\ell_1(x)$$

$$\ell_0(x) = \frac{x-(x_0+h)}{x_0-(x_0+h)}$$

$$\ell_1(x) = \frac{x-x_0}{x_0+h-x_0}$$

$$L_1(x) = \frac{f(x_0)(x-x_0-h)}{-h} + \frac{f(x_0+h)(x-x_0)}{h}$$

$$f'(x) = \frac{f(x_0)}{-h} + \frac{f(x_0+h)}{h} +$$

$$\left[\frac{(x-x_0)(x-x_0-h)}{2} f''(\varepsilon(x)) \right]$$

$$f'(x) = \frac{f(x_0+h) - f(x_0)}{h} + \frac{2(x-x_0)-h}{2}$$

$$f''(\varepsilon(x)) + \frac{(x-x_0)(x-x_0-h)}{2} \left[f''(\varepsilon(x)) \right]'$$

$x = x_0 \Rightarrow$ TPF

$$f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h} - \frac{h}{2} f''(\varepsilon(x))$$